COMPUTER NETWORK PROJECT

Group Members:

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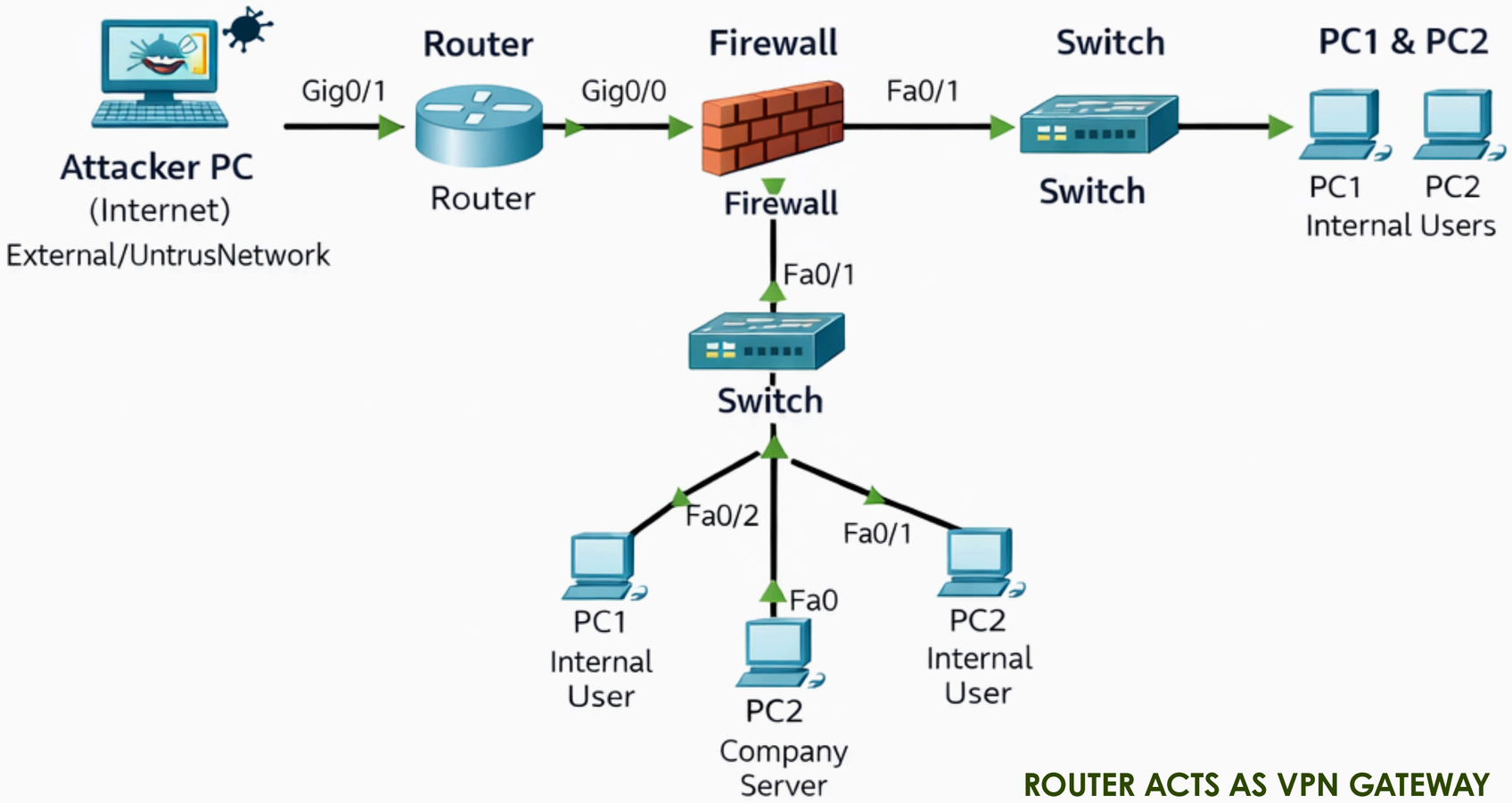
PROBLEM STATEMENT

- To design a secure network architecture for an organization
- To implement VPN for secure remote access
- To configure firewall policies for traffic control
- To deploy IPS for real-time threat detection and prevention
- To ensure confidentiality, integrity, and availability (CIA triad)

INTRODUCTION

- In today’s digital world, organizations face increasing cyber threats such as hacking, malware, and unauthorized access
- Securing network infrastructure is essential to protect sensitive data and ensure safe communication
- A strong security system requires multiple layers of protection
- This project focuses on implementing key security technologies like VPN, Firewall, and Intrusion Prevention System (IPS)
- These technologies work together to provide secure remote access, control network traffic, and prevent cyber attacks

BLOCK DIAGRAM



ROUTER ACTS AS VPN GATEWAY

FIREWALL + ACL SIMULATE IPS FUNCTIONALITY

METHODOLOGY

- Designed topology in Cisco Packet Tracer
- Configured IP addresses for all devices
- Connected attacker PC to external network
- Configured firewall for traffic filtering
- Applied ACL on router to simulate IPS
- Tested connectivity using ping

IMPLEMENTATION

- ACL created to block attacker IP
- Applied ACL on router interface
- Firewall configured to filter traffic
- Router acts as VPN Gateway (conceptual)

COMMANDS

Router Configuration commands

```
enable
configure terminal

interface g0/1
no shutdown
exit

interface g0/0
no shutdown
exit
```

Configure Router IP Addresses

Router needs two interfaces:

Outside (Internet side) - Connected to Attacker_PC.

```
interface g0/1
ip address 200.1.1.1 255.255.255.0
no shutdown
```

Inside (Company side) - Connected to Firewall.

```
interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
```

COMMANDS

Configure Attacker PC

IP Address: 200.1.1.2
Subnet Mask: 255.255.255.0
Gateway: 200.1.1.1

Configure Server

IP: 192.168.1.100
Subnet: 255.255.255.0
Gateway: 192.168.1.1

Configure Internal PCs

PC1
IP: 192.168.1.10
Subnet: 255.255.255.0
Gateway: 192.168.1.1

PC2
IP: 192.168.1.11
Subnet: 255.255.255.0
Gateway: 192.168.1.1

COMMANDS

Configure Firewall

enable
configure terminal

Configure inside interface:
interface vlan 1
nameif inside
security-level 100
ip address 192.168.1.254 255.255.255.0

Configure outside:
interface vlan 2
nameif outside
security-level 0
ip address 200.1.1.3 255.255.255.0

TESTING

Tests PC1 → Server communication.
ping 192.168.1.100

Test Router Connectivity
From PC1 Command Prompt type:
ping 192.168.1.1
This checks PC → Router.
If reply comes → internal network working.

Test Internet Side
Go to Attacker_PC
Desktop → Command Prompt
ping 200.1.1.1
This checks connection to Company_Router.
If reply comes → router outside interface working.

Test Firewall Protection
Test Attacker → Internal Network

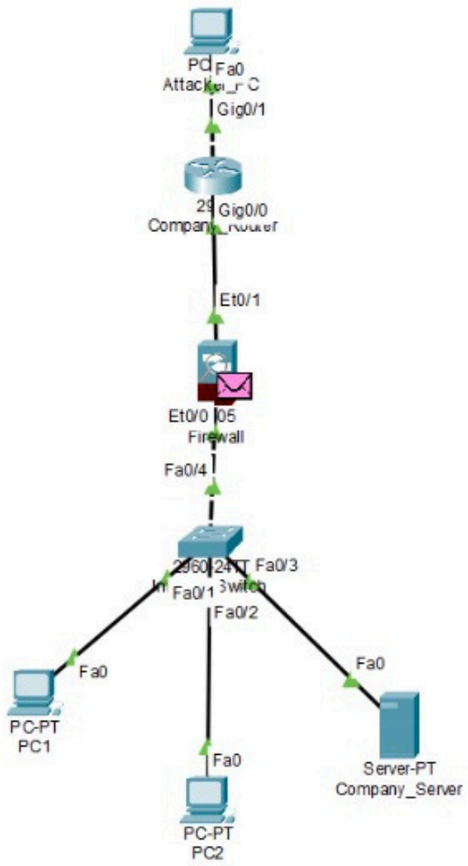
From Attacker_PC
ping 192.168.1.10
If firewall configured correctly
Ping will fail.
Request timed out
Meaning attacker cannot access company network.

RESULT

Cisco Packet Tracer - C:\Users\samiksha\Downloads\cnproject_vpn_firewall_ips.pkt

File Edit Options View Tools Extensions Window Help

Logical | Physical x: 496, y: 267 | Root



Simulation Panel

Event List

Vis.	Time(sec)	Last Device
	0.000	--
	0.001	Attacker_PC
	0.001	--
	0.002	Company_Router
	0.637	--
	0.638	Internal_Switch
	0.638	Internal_Switch
	0.638	Internal_Switch
	0.638	Internal_Switch
Visible	0.993	--

Reset Simulation | Constant Delay | Capt

Play Controls

Event List Filters - Visible Events

ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CD
DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, E
FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, IC
ICMPv6, IEC, IPSec, ISAKMP, IoT, IoT TCP, LA
LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP
OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED,
Profinet, RADIUS, REP, RIP, RIPng, RTP, SCC
SNMP, SSH, STP, SYSLOG, TACACS, TCP, T
Telnet, UDP, USB, VTP

Edit Filters | Show All/Nor

Time: 00:39:29.981 PLAY CONTROLS

Event List | Realtime | Simu

PT8200 4331 4321 IR8340 1941 2901 2911 819IOX 819HGW 829 1240 IR1101 PT-Router

829

Scenario 0

New | Delete

Toggle PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	Attac...	PC1	ICMP		0.000	N	0	(edit)	(delete)

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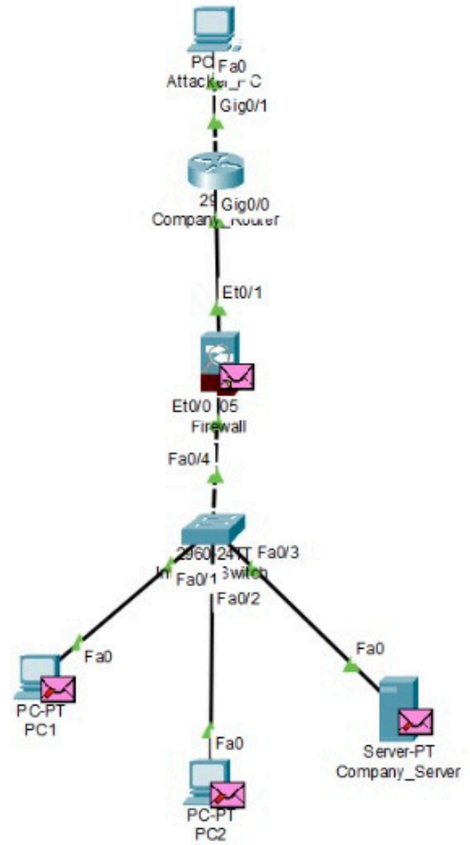
Cisco Packet Tracer - C:\Users\samiksha\Downloads\cnproject_vpn_firewall_ips.pkt

File Edit Options View Tools Extensions Window Help

Logical | Physical x: 858, y: 620

Root



Simulation Panel

Event List

Vis.	Time(sec)	Last Device
	0.003	Company_Server
	0.004	Internal_Switch
	0.643	--
	0.644	Internal_Switch
	0.644	Internal_Switch
	0.644	Internal_Switch
	0.644	Internal_Switch
	0.997	--
	0.998	Firewall
	2.643	--
Visible	2.644	Internal_Switch
Visible	2.644	Internal_Switch
Visible	2.644	Internal_Switch
Visible	2.644	Internal_Switch

Reset Simulation | Constant Delay | Captured to 2.644

Play Controls

Event List Filters - Visible Events

ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, Profinet, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters | Show All/None

Time: 00:39:43.626 PLAY CONTROLS:

Event List | Realtime | Simulation

PT8200 4331 4321 IR8340 1941 2901 2911 819IOX 819HGW 829 1240 IR1101 PT-Router

829

Scenario 0

New | Delete

Toggle PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	Company...	ICMP		0.000	N	0	(edit)	(delete)

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Search

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12:38 03-04-2026

CONCLUSION

- Secure network successfully implemented
- Firewall and ACL provide protection
- VPN Gateway ensures secure access
- IPS functionality simulated effectively

THANK YOU